COMPLETE RESEARCH RETROSPECTIVE

Latent-Space-Reasoning

Every research arc, theorem, validation, refutation, and open obligation of the program - from the founding denizen question through five complete theorem families and the dimension-asymptotic universality law.

> Six campaigns. Eleven adversarial reviews. 1085 ledger entries.

REPORT

PART 0 - WHAT THIS DOCUMENT IS AND WHY IT EXISTS

This is the sole surviving artifact of the Latent-Space-Reasoning research program. It contains every theorem with its full statement and proof, every validated empirical law with its numbers, every refuted claim with its cause, and every methodological lesson - written so that a reader with no access to the repository, the code, or the session history can reconstruct the entire state of knowledge.

The mission. Discover the general mathematics of high-dimensional spaces ($d \gg 1$): the laws that exist only when dimension is large. The axiom, held fixed throughout: mathematics behaves differently in high dimensions; concentration of measure, almost-orthogonality, interference, and distance uniformity are the native physics of these spaces. LLM latent spaces are one motivating instance, studied later once the foundation exists. The standing test every result had to pass: does this describe structure that R^n mathematics does not describe?

How the knowledge was produced. Pre-registered predictions frozen before every run; zero-free-parameter validation against derived formulas; an external adversarial reviewer (goal-aligned: attacking method and strategy to ensure the best path to the goal, never closing the mission) on a 2-hour cadence - 14 reviews total, every one catching at least one real defect; honest retraction recorded in place. Approximately 130 git commits and 1,085 append-only ledger rows document the provenance of everything below.

Status at pause (September 2026). Five complete theorem families; one dimension-asymptotic phase law (the universality crossover) confirmed across four dimensions; a frozen real-corpus prediction awaiting its confrontation; the program paused at the owner's request with all records current. Neither terminal condition (proven dead end, or reviewer-confirmed highest-tier contribution) has been reached.

PART I - OBJECTS AND DEFINITIONS (the vocabulary)

Everything below is stated in one shared notation. Features $f_1,\dots,f_k$ are vectors in $R^{(}$ (the 'superposed features'). Σ denotes a covariance matrix; $\mathrm{tr}(\cdot)$ the trace; $PR(\Sigma) = (\mathrm{tr}\Sigma)^2 / \mathrm{tr}(\Sigma^2)$ the participation ratio (effective dimension of the spectrum). k is the number of coexisting features; d the ambient dimension; $\gamma = n/d$ the overcompleteness of a frame $W \in R^{(d\times n)}$.

- A_2 (aggregate interference). $A_2 = \Sigma_{i<j} (f_i \cdot f_j)^2$ - the total squared crosstalk among superposed features. THE central observable of Part II.
- QSAE (quadratic superposition autoencoder). The loss $L(W) = \Sigma_i p_i \|W^TW e_i - e_i\|^2$ where p_i are usage weights (Zipf by default), e_i one-hot tokens, $W \in R^{(d\times n)}$ the code. Exactly: $L = \Sigma_i p_i(\|w_i\|^2-1)^2 + \Sigma_{i\neq j} p_i (w_i\cdot w_j)^2$.
- Softmax compression. Given a conditional kernel $P(j|i)$ (a row-stochastic matrix of task probabilities), minimize over rank-d factorized logits $z_j(i) = w_j\cdot h_i$ the cross-entropy $\Sigma_i w_i KL(P(\cdot|i) \| \mathrm{softmax}(z(\cdot,i)))$. The excess is this cost minus the conditional entropy floor.
- The logit residual. $\delta = z - \log q$ (model logits minus exact target logits). All compression geometry is a function of δ.
- The fourth-moment contraction operator. $C_i[x]$:= the map on symmetric matrices obtained by contracting the fourth-moment tensor of f_i against x on two slots. For Gaussian f_i with covariance Σ_i, Wick gives $C_i[x] = \mathrm{tr}(x\Sigma_i)\Sigma_i + 2\Sigma_i x \Sigma_i$. This operator is the central object of the general theorem (Part II.6).
- Frame spectrum. Eigenvalues of WW^T; the 'spectral exponent' α is the OLS slope of $\log \lambda_k$ vs $\log k$ - the quantity the field routinely measures.

PART II - THE THEOREMS (statements, proofs, verification status)

II.1 The sparse-exact optimum theorem

THEOREM / PROPOSITION 1 (sparse-exact optimum of the QSAE).

Let $G = W^TW$ (PSD, rank $\le d < n$) and $P = \mathrm{diag}(p)$. Then $L = \mathrm{tr}((G-I)P(G-I))$. Since G is PSD with rank $\le d$, it has an $(n-d)$-dimensional kernel on which $G-I$ acts as $-I$. Hence $L \ge \mathrm{tr}(P$ restricted to any $(n-d)$-dimensional subspace of ker $G)$, minimized by choosing ker G = span of the tail coordinates. The minimum is $L_{min} = \Sigma_{i>d}\, p_i$ (tail usage mass, S = the top-d usage set), attained by any orthonormal unit-norm storage $\{w_i\}_{i \in S}$ - for which the frame operator $WW^T = I_d$: a completely flat spectrum, condition number 1.

Proof. Write the loss exactly: $\|(G-I)e_i\|^2 = (G_{ii}-1)^2 + \Sigma_{j \ne i} G_{ji}^2$. Summing with weights p_i gives $L = \mathrm{tr}((G-I)P(G-I))$. Now $G \succeq 0$ with rank $\le d$ means G has $n-d$ zero eigenvalues; on that kernel K, $(G-I)$ restricted equals $-I$, so the contribution is $\mathrm{tr}(P|_K) = \Sigma_{i \in K} p_i$. Choosing K to carry the smallest weights (the usage tail) minimizes this part, and the stored block contributes zero. □

Verification. L-BFGS from 5 independent inits at six (s, n, d) configurations: measured loss minus the tail-mass bound = 0.000000 at all six; nonzero-norm count = d exactly; spectrum at the optimum flat. Consequence: power-law spectra cannot come from the quadratic objective's optimum - the optimum anti-mixes (organizes orthogonality) rather than mixes.

II.2 The exact isotropic interference theorem

THEOREM / PROPOSITION 2 (isotropic A_2: exact mean and variance).

For independent unit-norm isotropic features and the energy readout, $A_2 = 2\Sigma_{i<j} W_{ij}$ with $W_{ij} = (f_i \cdot f_j)^2$. Exactly: $E[A_2] = k(k-1)/d$; per-pair $E[W] = 1/d$, $\mathrm{Var}(W) = 3/(d(d+2)) - 1/d^2$; all covariances between distinct pair-squares vanish identically (shared-index pairs: $E[W_{ij} W_{il}] = (1/d)^2 = E[W]^2$ by conditional independence and $\|f_i\| = 1$ exactly; disjoint pairs: independent). Hence $\mathrm{Var}(A_2) = 2k(k-1)(2d-2)/(d^2(d+2))$ and cv $\approx 2/k$. By Chebyshev, $A_2 \in [E/2, 2E]$ with probability $\ge 1 - 8/k^2$: two-phase frontier $k^* = \Theta(\sqrt{d})$ - mean-field (independent-feature) computation valid below, aggregate interference inescapable above.

Verification. Pre-registered simulation: crossing confirmed (A_2 = 0.87-0.97 at k = √d; 0.04-0.05 at √d/4; 14.8-15.7 at 4√d), concentration as predicted. The variance constant was corrected once (an ordered/unordered double-count caught by adversarial review; conclusions unchanged).

II.3 The participation-ratio law

THEOREM / PROPOSITION 3 (anisotropic frontier).

For unit-energy features with common covariance Σ ($E[ff^T] = \Sigma/\mathrm{tr}\Sigma$): per-pair interference $E[(f_i \cdot f_j)^2] = \mathrm{tr}(\Sigma^2)/(\mathrm{tr}\Sigma)^2 = 1/PR(\Sigma)$ exactly. Therefore $E[A_2] = k(k-1)/PR$ and aggregate interference reaches order one at $k^* = \Theta(\sqrt{PR(\Sigma)})$. For power-law spectra $\lambda_l \propto l^{-\alpha}$ with $1/2 < \alpha < 1$: $PR = \Theta(d^{2-2\alpha})$ and $k^* = \Theta(d^{1-\alpha})$ - a steeper trained spectrum taxes computation by $d^{\alpha-1/2}$ relative to isotropy. (At $\alpha = 1$ the scale is $\Theta(\log d)$; $\alpha > 1$ gives $\Theta(1)$.)

Proof. $E[(f_i \cdot f_j)^2]$ with $f = (\mathrm{tr}\Sigma/d)^{-1/2}\, \Sigma^{-1/2} u$ independent: $= (d/\mathrm{tr}\Sigma)^2\, \mathrm{tr}(\Sigma\, E[uu^T]\, \Sigma\, E[uu^T]) = (d/\mathrm{tr}\Sigma)^2\, \mathrm{tr}(\Sigma \cdot \Sigma/d) \cdots = \mathrm{tr}(\Sigma^2)/(\mathrm{tr}\Sigma)^2$. Checks: isotropic $\Sigma = I$ gives $1/d$ (Theorem 2); rank-1 Σ gives 1 (parallel features maximally interfere). □

Verification. Per-pair law exact to 0.4-2.4% at $\alpha \le 0.5$; crossings bracketed 8/8 cells. Scope (adversarially enforced): expectation-level relation for the specified quadratic model; not a decoder/task capacity theorem; high-probability frontier proved only isotropic.

II.4 Covariance insufficiency and the localization law

THEOREM / PROPOSITION 4 (covariance does not determine sharpness).

Two ensembles with identical Σ - Gaussian directions vs coordinate-sparse directions (u uniform over $\pm\sqrt{d}\, e_r$) - share the same $E[A_2]$ to within a few percent but differ in cv by 21× (measured: 1.64 vs 0.076 at $d = 512$, $k = 64$, $\alpha = 0.5$) and in spread (p90/p10) by 14×. For coordinate-sparse directions on a diagonal $\Sigma = \mathrm{diag}(\sigma_r)$: $cv^2 \approx 2(\Psi_1 - 1)/k^2 + 2(\Psi_2 - 1)/k$ with $\Psi_1 = d^2\mathrm{tr}(\Sigma^4)/(\mathrm{tr}\Sigma^2)^2$ and $\Psi_2 = d\,\mathrm{tr}(\Sigma^4)/(\mathrm{tr}\Sigma^2)^2$ - exact at $\alpha = 0$ in both terms ($\sqrt{(2d)/k}$ law reproduced to 2%), no free constants. Gaussian directions collapse both functionals to the clean $2/k$ law.

Meaning. The spectrum sets the frontier's mean; the direction distribution's localization sets its sharpness. Any second-moment tool (the entire R^n toolkit) is blind to the second structure.

II.5 The exact birthday covariance and limit law

THEOREM / PROPOSITION 5 (fixed-k multinomial covariance).

For $S = \Sigma_r w_r C(N_r, 2)$ with $N \sim$ Multinomial(k; uniform p = 1/d), falling factorials $(k)_m$: $E[S] = (1/2)(k)_2 p^2 W_1$ and $Var(S) = (1/4)[W_2 V + (W_1^2 - W_2) C]$ with $V = (k)_4 p^4 + 4(k)_3 p^3 + 2(k)_2 p^2 - (k)_2^2 p^4$ and $C = -(k)_2(4k-6) p^4$ (the NEGATIVE fixed-k correlation - the term Poissonization misses). $W_1 = \Sigma w_r$, $W_2 = \Sigma w_r^2$. Verification: 12/12 cells at ratios 0.76-1.09, zero free parameters. Law class: governed by $\lambda^2 \cdot PR_W$ ($\lambda = k/d$ density, PR_W the weight participation ratio) in the dense regime; asymptotic normality follows from the occupancy sum-function CLT (Holst 1972; Karlin 1967) when the Lindeberg/max-cell condition holds; compound-Poisson near the birthday threshold.

The refutation that preceded it: the Poissonized (independent-channel) prediction was pre-registered and failed 2/12 - measured cv^2 dropped ~10× per 4× λ while the Poissonized formula is λ-free. The fixed-count conditioning is load-bearing, not a le-Cam correction.

II.6 The general fourth-moment theorem (HP-G)

THEOREM / PROPOSITION 6 (interference moments via the contraction operator).

For arbitrary independent features with covariances Σ_i (triangular array): (i) $E[A_2] = \Sigma_{i \neq j} tr(\Sigma_i \Sigma_j)$ - second moments only. (ii) The variance is determined by Σ and the fourth-moment contraction operators C_i through: shared-index covariance = $2\, tr(\Sigma_j \Sigma_i \Sigma_l \Sigma_i)$ (Gaussian instance; general: Σ_j, $C_i[\Sigma_l]$ minus product of means), and pair-variance $E[(f_i f_j)^4] - E[(f_i f_j)^2]^2$. Homogeneous Gaussian assembly: $Var(A_2) = C(k,2)[2\, tr(\Sigma^2)^2 + 6\, tr(\Sigma^4)] + 4k\, C(k-1,2)\, tr(\Sigma^4)$, giving the explicit L^2 criterion $cv^2 = [2 + 6 \cdot t_4/t_2^2]/C(k,2) + 4kC(k-1,2) \cdot t_4/[C(k,2)^2 t_2^2]$, $t_2 = tr(\Sigma^2)$, $t_4 = tr(\Sigma^4)$. Certifications: Wick instance 0.989; pair term 1.0032; counting 0.9994/1.0021/1.0031 by brute force; the corrected criterion matches stored data (0.0673 vs 0.0708).

THEOREM / PROPOSITION 7 (the universality crossover law).

The Gaussian universality class of the interference statistic has a dimension-dependent stability boundary: for mixture directions $f = \sqrt{(1-\eta)}$ Gaussian + $\sqrt{\eta}$ coordinate-sparse, the class boundary obeys $\eta^*(d) \sim d^{-1/2}$ - departure when $\eta^2 d \sim 1$ (the localized component's fourth-moment contraction, of mass $\sim \eta \cdot d$, reaches unity). Confirmed by pre-registered bracketing: eta in (0.30,0.35] at d = 512; (0.15,0.25] at d = 2048 and 8192; and the pinning cell at d = 32768 landing exactly in its predicted bracket (exceedance at $\eta = 0.12$: cv 0.0637 vs line 0.0630; within at $\eta = 0.08$: 0.0389). Mechanism: the sparse operator's diagonal probe $R = \Sigma_a E[f_a^4]/\Sigma_a (E f_a^2)^2$ equals 3.0 for Gaussian (dimension-free) vs ~d for sparse (dimension-growing). The formal Lipschitz proof that the operator bookkeeping reproduces $d^{-1/2}$ exactly is written as a skeleton and remains open.*

II.7 The compression theorems (the Orlicz arc)

THEOREM / PROPOSITION 8 (exact CGF identity).

For any target distribution q and logits z with residual $\delta = z - \log q$: $KL(q \,\|\, softmax(z)) = \log E_q[e^{\delta}] - E_q[\delta]$ - the cumulant generating function of the logit residual under q. Series: $KL = \Sigma_{m \geq 2} \kappa_m(\delta)/m!$. The Fisher (quadratic) compression law is exactly the κ_2 term.

Proof. KL(q‖s) = −H(q) − q, z + logΣ e^z; substituting z = log q + δ and collecting: = log E_q[e^θ] − E_q[δ] identically. □

THEOREM / PROPOSITION 9 (measured qualitative argmin change).

At steep usage (Zipf s = 1) far below representability, the compression optimum settles (12 seeds within 0.0005) at a solution whose quadratic component alone (0.0966) is 7× the Fisher-optimal value (0.0139) and exceeds the true total (0.0859), with $\kappa_3 = -0.005$ partially cancelling: the optimum deliberately accepts 7× the variance to buy negative-skew cumulant structure. Softmax compression is a different mathematical object from low-rank approximation - the argmin changes qualitatively, not perturbatively. Variance is nearly irrelevant (envelope degeneracy below); one-sided exponential moments govern.

THEOREM / PROPOSITION 10 (envelope degeneracy).

The moment problem 'minimize KL over residual distributions with fixed variance' is degenerate: rare huge negative jumps (probability $\varepsilon \to 0$, magnitude $\sim \sqrt{(v/\varepsilon)}$) carry the variance while $E_q[e^{\delta}] \to 1$, so inf KL = 0 at any fixed variance. Polynomial moments of δ do not govern KL compression; the one-sided exponential moment does (Orlicz scale).

THEOREM / PROPOSITION 11 (duality impossibility cluster).

Three Eckart-Young-style duals fail structurally: (a) the SVD-based dual has an unbounded inner problem (inf of a linear functional on the unconstrained rank-d variety is $-\infty$); (b) the Donsker-Varadhan dual inherits that unboundedness; (c) the λ-regularized dual is vacuous at any λ that preserves the problem (the optimum carries $\|R\|^2 \sim 1.1 \times 10^4$). Root cause: KL geometry has no norm ball - logits scale freely because only entropy differences matter.

THEOREM / PROPOSITION 12 (first-order structure; verified).

The optimal rank-d R of the compression problem satisfies the Eckart-Young-type orthogonality: the weighted gradient field $W_b(softmax(r^*_b) - q_b)$ vanishes ON the row space of R* (lives entirely in the complement block) - verified at 1.0000*

(off-space gradient energy fraction) at the settled optimum. Strict local minimality: Lanczos Hessian on the GL(d)-quotient gives $\lambda_min = 1.18\times10^{-5} > 0$ - positive in every computation but at the finite-difference noise boundary; evidence-grade. (The naive full-space certificate was challenged for gauge modes; the corrected quotient computation supersedes it.)

THEOREM / PROPOSITION 13 (hierarchical family).

For the level-mixture kernel $P(j|i) = (1-\rho)p_j + \rho \Sigma_\ell \pi_\ell 1[\text{same level-}\ell\text{ ancestor}]$: (i) the centered log-kernel has rank exactly $2^L - 1$ (nested-bump orthogonality: same-level bumps disjoint; cross-level bumps orthogonal after centering by the tree property); (ii) bundle-scale law $\sigma_\ell \sim \sqrt{(m_\ell)\Delta_\ell}$ (level mass × level logit gap), geometric under geometric π; (iii) the excess curve is a recursive double tail (level envelope × within-bundle Zipf block-mass tails); (iv) empirically: smooth log-convex decay to EXACT zero at the representability rank (measured, both ρ arms), Fisher envelope tight 1.08-1.28× far below representability, super-quadratic efficiency near it; (v) the steeper-π probe is decisive: the family cannot reach the CGF regime at any steepness - the shared-root floor bounds likelihood contrasts (construction-forced, scoped as such).

II.8 Supporting identities (from earlier campaigns, retained)

- Write-orbit bound (CR-15): the write orbit from one initial place is bounded by $\prod_r(|V_r|+1)$; complete initialization gives exactly $\prod_r|V_r|$ places; η-accurate transition-closed realizations bound equal-endpoint histories by 2η under every registered continuation; a finite behavioral budget N bounds lifetime error by $(N-1)\varepsilon$ - sharp.
- Quadratic interference hierarchy (p = 2 instance): for isotropic features and normalized fixed M, 1-body terms are O(1) and 2-body terms $O(d^{-1/2})$ RMS $(E[(f^{T}Mf')^2] = \|M\|^2/d^2)$ - the HD analogue of the cluster expansion; the general-t conjecture $d^{-(t-1)/2}$ remains open (and the naive general-t eigenvalue law was REFUTED: the correct per-term law is $\|T\|^2/d^t$, each vector contributing 1/d).
- Rate-distortion benchmark: the scalar allocation problem $\min \sigma^2\Sigma\, p_i/\lambda_i$ s.t. $\Sigma\lambda = P$ has optimizer $\lambda^* \propto \sqrt{p_i}$ (exponent s/2) - retained as a labeled benchmark only; by Theorem 1 it cannot describe quadratic-loss equilibria, and by the gauge lesson it cannot be compared to factor spectra without fixing a gauge.
- Excess-CE truncation bound: $KL(q\|\text{softmax}(\hat{z})) \le 2\cdot\max_j|z^*_j - \hat{z}_j|$ (via 1-Lipschitz logsumexp): compression quality is governed by worst-row logit deviation.

PART III - THE EMPIRICAL LAWS (numbers a reader can use)

III.1 Measurement validity: the estimator-manufacture findings

 The standard spectral exponent (OLS of log λ_k vs log k over the top-90% ranked eigenvalues) has (overcompleteness γ, profile β)-dependent bias up to ±0.2 for TRUE power-law inputs - γ = 4 inflates shallow exponents; γ = 16 deflates steep ones. Bias is asymptotic in d by d ~ 128; it is a function of (γ, β) only.

 FLAT inputs read as α ≈ 0.40-0.45 at γ = 4 (threshold 0.35 fails); at γ = 16, 0.20-0.23 - matching the record's 'flat init ≈ −0.25' readings at real-model overcompleteness.

 Bulge (non-power-law) inputs manufacture steep slopes (α̂ up to ~12) but a quadratic-in-log alternative correctly rejects them (ΔR^2 = 0.13-0.28 in favor of curvature). Diagnostic battery: window sensitivity {top-50%, top-90%, full} + curved alternative.

 Consequence: spectral exponents are comparable only within fixed (γ, window). Any cross-scale exponent comparison in the earlier literature of this program (and plausibly elsewhere) carries this bias.

III.2 The rank-2 collapse (softmax on unigram Zipf)

Tied-embedding softmax on independent Zipf(s=1) tokens (n = 1024, d = 64, Adam, 192k steps, 3 seeds): top-2 eigenvalue share ≥ 0.984 at EVERY checkpoint from t = 3k; token norms uniformize (spread 1.9 → 1.4); curved fits beat power law at 12/12 checkpoint-seeds (R^2 of power-law fit only 0.54-0.71); measured α the norms-only control (0.22). The fitted slopes are directional structure (the collapse coherence - common direction + frequency direction), not norm allocation. The theoretical construction: w_i = u + α_i e with u·e = 1, α_i log p_i, ‖u‖ → ∞ realizes the target exactly in rank 2.

III.3 The compression two-regime structure

Planted-block kernel (B blocks, mixture ρ, Zipf s), measured optimal excess vs the Fisher prediction - across B {8, 32}, ρ {0.3, 0.7}, s {0.5, 1}: tight (1.08-1.28×) far below representability at s = 0.5; large-deviation underestimate 1.86-2.79× at s = 1 far below; super-quadratic efficiency (optimizer beats second-order) approaching the knee, monotone in d/B, down to 0.02-0.17×. The s = 1 gap is genuine (not optimizer error: 12 seeds within 0.0005). Regime structure recurs identically on the hierarchical family and (partially) on the real corpus confrontation.

III.4 The universality-crossover measurements

Mixture ensembles √(1−η)Gaussian + √η coordinate-sparse, k = 64, exceedance rule cv > 2×Wick-prediction: at η = 0.4 measured cv grows 0.059 → 0.088 → 0.155 across d = 128/512/2048 (directional confirmation); fine grids bracket η*(512) (0.30,0.35], η*(2048), η*(8192) (0.15,0.25]; pinning cell d = 32768: 0.08 within / 0.12 exceeds - the d^−1/2 bracket. Gaussian-class robustness at η = 0.1 (indistinguishable from η = 0) then sharp departure by η = 0.5.

III.5 Earlier-program durable empirics

 Perturbation hierarchy (86 experiments, 10 models): position >> operator family >> sign >> answer identity >> meaning - scoped to tested panels; meaning undetected and confounded with answer change in arithmetic.

 Multiplicative sensitivity: d[cell, perturbation] ≈ fragility × magnitude, log-space R^2 0.82-0.88 (later scoped as first-order-Taylor consistency, with the content in the 12-25% residual).

 Depth-dependent suffix effect: one fixed appended comment raises correct-value probability up to +12.6 pp at nesting depth 4 (one model family; mechanism unresolved - lexical cueing, added computation, and attention-state change not separated).

 Cross-model asymmetry: perturbation-TYPE transfer ~40% shared across models; cell-FRAGILITY transfer ~0% - rules about change types may generalize; per-input vulnerability maps do not.

PART IV - THE FAILURE REGISTER AND THE LESSONS (as valuable as Part II)

Belief that died	How it died	LESSON (transferable)
'Zipf + gradient descent produces power-law spectra' (RESOLVED)	'Zipf + gradient descent produces power-law spectra' (RESOLVED)	'Zipf + gradient descent produces power-law spectra' (RESOLVED)
Poissonized occupancy prediction for weighted birthday statistics	Poissonized occupancy prediction for weighted birthday statistics	Poissonized occupancy prediction for weighted birthday statistics
Variance-KL discount law v/2 − c·v^{3/2}	Variance-KL discount law v/2 − c·v^{3/2}	Variance-KL discount law v/2 − c·v^{3/2}
Three duality certificates (SVD, unconstrained DV, regularized)	Three duality certificates (SVD, unconstrained DV, regularized)	Three duality certificates (SVD, unconstrained DV, regularized)
'Capacity bridge' from the participation-ratio law (4 wording instances)	'Capacity bridge' from the participation-ratio law (4 wording instances)	'Capacity bridge' from the participation-ratio law (4 wording instances)
Norm-fluctuation order-of-magnitude exhibit	Norm-fluctuation order-of-magnitude exhibit	Norm-fluctuation order-of-magnitude exhibit
HP-F falsifier sweep (designed, not run)	HP-F falsifier sweep (designed, not run)	HP-F falsifier sweep (designed, not run)
'Disjointness gates the CGF regime' (general claim)	'Disjointness gates the CGF regime' (general claim)	'Disjointness gates the CGF regime' (general claim)
Corpus spectral-inheritance matching (predict embedding spectra from co-occurrence spectra)	Corpus spectral-inheritance matching (predict embedding spectra from co-occurrence spectra)	Corpus spectral-inheritance matching (predict embedding spectra from co-occurrence spectra)
11.7× 'strict local minimum CERTIFIED' (Lanczos)	11.7× 'strict local minimum CERTIFIED' (Lanczos)	11.7× 'strict local minimum CERTIFIED' (Lanczos)
Hierarchical staircase-knee prediction	Hierarchical staircase-knee prediction	Hierarchical staircase-knee prediction
Synthetic-family tunnel (meta-failure, diagnosed 3×)	Synthetic-family tunnel (meta-failure, diagnosed 3×)	Synthetic-family tunnel (meta-failure, diagnosed 3×)

The meta-lesson running through all twelve: the discipline worked. Every failure above was caught by pre-registration, by validation data, by an adversarial reviewer, or by a normalization assertion - before it could contaminate the success register. Nothing in Part II has been retracted.

PART V - STATE, OPEN PROBLEMS, AND THE STANDING JUDGMENT

V.1 Open problems, in priority order

 The frozen confrontation. The Fisher-tail excess-CE curve for the real SmolLM2-tokenized WikiText co-occurrence kernel (120M tokens, usage-probability-weighted, d grid 8-190) was committed BEFORE measurement. The measuring run completed one cell (d = 16: measured excess 6.01 vs full-kernel curve 1.24 - uninterpreted: the honest comparison needs the subsample's own reference tail, computed at run end, not reached). Paused mid-run by the owner. Scoring rule frozen: direct curve comparison, no refitting.

 G4-formal. The Lipschitz bookkeeping proving $\eta^* \sim d^{-1/2}$ from the operator perturbation (skeleton written: both variance terms are linear in C_i; the mixture perturbation carries $\eta(1-\eta)\cdot d$ diagonal-probe mass; the k-power cancellation must reproduce the measured law).

 Universality-class separation. Construct two arrays with identical second AND fourth moments but different limits of A_2 (or prove impossible); anchor the counterexample side of the classification.

 Corpus replication. Same frozen-test protocol on the qwen3 and phi-1.5 kernels (assets staged: 120M tokens each, ±2-window co-occurrences as CSR, ~9.5M nonzeros each).

 The bridge back. Intervene on a real model's representation spectrum and measure its superposition capacity against the $d^{(1-\alpha)}$ frontier - the first in-model test of Part II.3.

V.2 What is honestly NOT established

 No theorem here has been tested by intervention in a live transformer; all empirics are synthetic kernels, random ensembles, or (in the paused test) corpus statistics.

 Every theorem is scoped to its specification: quadratic readouts, softmax factorizations, planted/hierarchical kernels, specific ensembles. None is a claim about transformer residual streams.

 Several claims are expectation-level (Theorem 3), evidence-grade (Theorem 12 strictness), or empirically established with mechanism but without formal proof (Theorem 7's exact constant).

 All review was internal (an external CLI reviewer), not independent mathematical peer review.

V.3 The standing judgment

The direction is alive and productive. Five theorem families stand with validated or certified cores; the universality crossover is a genuine dimension-asymptotic phase law of the kind the mission demanded - it does not exist at low dimension at all. The program's distance to its own terminal conditions (proven dead end, or highest-tier confirmed contribution) remains large. The next decisive event is the confrontation between the compression theory and a real corpus it did not design. The operating discipline - pre-registration, zero-parameter validation, goal-aligned adversarial review, honest retraction - is itself a transferable artifact: it caught every error in Part IV before anything false entered Part II.